Hypothesis Testing

A statistical hypothesis test is a method of statistical inference used to decide whether the data at hand sufficiently support a particular hypothesis. Hypothesis testing allows us to make probabilistic statements about population parameters.

Null and Alternate Hypothesis

1. Null hypothesis (H0):
In simple terms, the null hypothesis is a statement that assumes there is no significant effect or relationship between the variables being studied. It serves as the starting point for hypothesis testing and represents the status quo or the assumption of no effect until proven otherwise. The purpose of hypothesis testing is to gather evidence (data) to either reject or fail to reject the null hypothesis in favour of the alternative hypothesis, which claims there is a significant effect or relationship.

2. Alternative hypothesis (H1 or Ha):
The alternative hypothesis, is a statement that contradicts the null hypothesis and claims there is a significant effect or relationship between the variables being studied. It represents the research hypothesis or the claim that the researcher wants to support through statistical analysis.

Performing a Z test Example 1

Suppose a company is evaluating the impact of a new training program on the productivity of its employees. The company has data on the average productivity of its employees before implementing the training program. The average productivity was 50 units per day with a known population standard deviation of 5 units. After implementing the training program, the company measures the productivity of a random sample of 30 employees. The sample has an average productivity of 53 units per day. The company wants to know if the new training program has significantly increased productivity.

Example 2

Suppose a snack food company claims that their Lays wafer packets contain an average weight of 50 grams per packet. To verify this claim, a consumer watchdog organization decides to test a random sample of Lays wafer packets. The organization wants to determine whether the actual average weight differs significantly from the claimed 50 grams. The organization collects a random sample of 40 Lays wafer packets and measures their weights. They find that the sample has an average weight of 49 grams, with a known population standard deviation of 4 grams.

T-tests

A t-test is a statistical test used in hypothesis testing to compare the means of two samples or to compare a sample mean to a known population mean. The t-test is based on the t-distribution, which is used when the population standard deviation is unknown and the sample size is small.

There are three main types of t-tests:

One-sample t-test: The one-sample t-test is used to compare the mean of a single sample to a known population mean. The null hypothesis states that there is no significant difference between the sample mean and the population mean, while the alternative hypothesis states that there is a significant difference.

Independent two-sample t-test: The independent two-sample t-test is used to compare the means of two independent samples. The null hypothesis states that there is no significant difference between the means of the two samples, while the alternative hypothesis states that there is a significant difference.

Paired t-test (dependent two-sample t-test): The paired t-test is used to compare the means of two samples that are dependent or paired, such as pre-test and post-test scores for the same group of subjects or measurements taken on the same subjects under two different conditions. The null hypothesis states that there is no significant difference between the means of the paired differences, while the alternative hypothesis states that there is a significant difference.

Single Sample t-test

A one-sample t-test checks whether a sample mean differs from the population mean.
Assumptions for a single sample t-test
1. Normality - Population from which the sample is drawn is normally distributed
2. Independence - The observations in the sample must be independent, which means that the value of one observation should not influence the value of another observation.
3. Random Sampling - The sample must be a random and representative subset of the population.
4. Unknown population std - The population std is not known.

Suppose a manufacturer claims that the average weight of their new chocolate bars is 50 grams, we highly doubt that and want to check this so we drew out a sample of 25 chocolate bars and measured their weight, the sample mean came out to be 49.7 grams and the sample std deviation was 1.2 grams. Consider the significance level to be 0.05

F - Distribution

Continuous probability distribution: The F-distribution is a continuous probability distribution used in statistical hypothesis testing and analysis of variance (ANOVA).

Fisher-Snedecor distribution: It is also known as the Fisher-Snedecor distribution, named after Ronald Fisher and George Snedecor, two prominent statisticians.

Degrees of freedom: The F-distribution is defined by two parameters - the degrees of freedom for the numerator (df1) and the degrees of freedom for the denominator (df2).

Positively skewed and bounded: The shape of the F-distribution is positively skewed, with its left bound at zero. The distribution's shape depends on the values of the degrees of freedom.

Testing equality of variances: The F-distribution is commonly used to test hypotheses about the equality of two variances in different samples or populations.

Comparing statistical models: The F-distribution is also used to compare the fit of different statistical models, particularly in the context of ANOVA.

F-statistic: The F-statistic is calculated by dividing the ratio of two sample variances or mean squares from an ANOVA table. This value is then compared to critical values from the F-distribution to determine statistical significance.

Applications: The F-distribution is widely used in various fields of research, including psychology, education, economics, and the natural and social sciences, for hypothesis testing and model comparison.

One way ANOVA test

One-way ANOVA (Analysis of Variance) is a statistical method used to compare the means of three or more independent groups to determine if there are any significant differences between them. It is an extension of the t-test, which is used for comparing the means of two independent groups. The term "one-way" refers to the fact that there is only o ne independent
variable (factor) with multiple levels (groups) in this analysis.

The primary purpose of one-way ANOVA is to test the null hypothesis that all the group means are equal. The alternative hypothesis is that at least one group mean is significantly different from the others.

Steps

- Define the null and alternative hypotheses.
- Calculate the overall mean (grand mean) of all the groups combined and mean of all the groups individually.
- Calculate the "between-group" and "within-group" sum of squares (SS).
- Find the between group and within group degree of freedoms
- Calculate the "between-group" and "within-group" mean squares (MS) by dividing their respective sum of squares by their degrees of freedom.
- Calculate the F-statistic by dividing the "between-group" mean square by the "within-group" mean square.

Source of Variation	Sum of Squares	Degrees of Freedom	Mean Squares (MS)	F
Within	$SSW = \sum_{j=1}^{k}\sum_{j=1}^{l}(X-\overline{X}_j)^2$	$df_w = k - 1$	$MSW = \frac{SSW}{df_w}$	$F = \frac{MSB}{MSW}$
Between	$SSB = \sum_{j=1}^{k}(\overline{X}_j - \overline{X})^2$	$df_b = n - k$	$MSB = \frac{SSB}{df_b}$	
Total	$SST = \sum_{j=1}^{n}(\overline{X}_j - \overline{X})^2$	$df_t = n - 1$		

•Calculate the p-value associated with the calculated F-statistic using the F-distribution and the appropriate degrees of freedom. The p-value represents the probability of obtaining an F-statistic as extreme or more extreme than the calculated value, assuming the null hypothesis is true.

• Choose a significance level (alpha), typically 0.05.
• Compare the calculated p-value with the chosen significance level (alpha).

a. If the p-value is greater than alpha, fail to reject the null hypothesis, concluding that there is not enough evidence to suggest a significant difference between the group means.

b. It's important to note that one-way ANOVA only determines if there is a significant difference between the group means; it does not identify which specific groups have significant differences. To determine which pairs of groups are significantly different, post-hoc tests, such as Tukey's HSD or Bonferroni, are conducted after a significant ANOVA result.

PART 1 – One Sample Z-Test (Mean)

Task 1: One Sample Z-Test (Population Std Known)

A hospital claims that the **average cancer treatment cost** is **$50,000**.

From your dataset, take a random sample of 40 patients.

Given:

- Sample mean = $53,200
- Population standard deviation = $8,000
- n = 40
- α = 0.05

Task:

1. Test whether the true mean treatment cost is different from $50,000.
2. Use **Z-test**
3. Calculate **Z-statistic**
4. Find **p-value**
5. Make decision using p-value
6. Write business conclusion

PART 2 – One Sample T-Test

Task 2: One Sample T-Test (Population Std Unknown)

A research study claims that average **Survival Years = 5 years**.

From dataset:

- n = 25

- Sample mean = 4.3
- Sample std = 1.8
- $\alpha = 0.05$

Task:

1. Test if survival years are significantly different from 5.
2. Use **One Sample T-Test**
3. Calculate T-statistic
4. Find p-value
5. Decision using p-value
6. Write healthcare interpretation

PART 3 – Independent Two Sample T-Test

Task 3: Smoking vs Non-Smoking Severity

You want to check whether **Smoking affects Severity Score**.

Group 1 (Smokers):

- n1 = 30
- mean1 = 6.1
- std1 = 1.4

Group 2 (Non-Smokers):

- n2 = 30
- mean2 = 5.2

- std2 = 1.6

$\alpha = 0.05$

Task:

1. Test if mean severity differs between groups.
2. Use **Independent Two Sample T-Test**
3. Calculate pooled variance
4. Calculate T-statistic
5. Find p-value
6. Write interpretation in medical terms

PART 4 – Paired T-Test

Task 4: Before vs After Treatment

You measure severity score before and after treatment for same 20 patients.

Mean difference (Before - After) = 1.2
Std of difference = 1.5
$n = 20$
$\alpha = 0.05$

Task:

1. Test whether treatment significantly reduces severity.
2. Use **Paired T-Test**
3. Calculate T-statistic
4. Find p-value
5. Make decision

6. Write hospital management conclusion

PART 5 – Z-Test for Proportion

Task 5: Stage IV Proportion Test

Hospital claims that **20% patients are Stage IV**.

From sample:

- n = 200
- Stage IV patients = 52

α = 0.05

Task:

1. Test whether actual proportion differs from 20%.
2. Use **Z-test for proportion**
3. Calculate p-hat
4. Calculate Z-statistic
5. Find p-value
6. Give conclusion

🔷 PART 6 – One Way ANOVA

Task 6: Cancer Type vs Treatment Cost

You want to check whether **mean Treatment Cost differs across Cancer Types**:

Cancer Type	Sample Size	Mean Cost	Std
Lung	30	62,000	7,000
Breast	30	54,000	6,000
Colon	30	58,000	6,500

$\alpha = 0.05$

Task:

1. Define hypotheses
2. Calculate SST, SSW
3. Calculate F-statistic
4. Find p-value
5. Decision using p-value
6. Business interpretation

BONUS – Two Way ANOVA (Advanced)

Test impact of:

- Cancer Type
- Gender
  On Treatment Cost

Check:

- Main effect of Cancer Type
- Main effect of Gender
- Interaction effect

What Students Must Submit

For every test:

1. H_0 and H_1
2. Test Statistic formula
3. Calculated value
4. p-value
5. Decision
6. Plain English interpretation